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# ELIZABETH ARYSLANOVA

## MACHINE LEARNING ENGINEER

### SUMMARY

Freelance machine learning engineer with a master's degree in optics and a background in mathematical modeling. With 17 years of dedicated mathematics teaching experience and a successful track record in online course production and consulting on 20+ projects.

### EDUCATION

#### POSTGRADUATE DEGREE (Unfinished)

Mathematical Modeling,  
Numerical Methods and  
Software Solutions  
ITMO University  
2014 - 2018

#### M.Sc, B.Sc.OF ENGINEERING AND TECHNOLOGY

Photonics and Optical  
Information Technology  
ITMO University  
2008 - 2014

### SKILLS

#### Programming & Frameworks

Python, PyTorch, Librosa,  
NumPy, SciPy, Matplotlib,  
PyQt5, PySide6, HTML, CSS

#### Software

Logic Pro X, Wolfram  
Mathematica, Final Cut,  
DaVinci Resolve, Affinity Photo,  
Affinity Designer,

#### Other

Capturing visuals,  
Public speaking, Teaching,  
Simplifying complexity,  
Soldering, Audio recording

### LANGUAGES

English, Russian - Native  
German - Intermediate

### PROFESSIONAL EXPERIENCE

#### MACHINE LEARNING ENGINEER

Freelance

Remote | Feb 2024 - Present

- Contributed to the development of **10+ commercial audio plugins** by designing and training machine learning models for audio analysis and intelligent audio processing.
- Designed and evaluated deep learning models (CNNs, TCNs, and RNNs) for audio analysis..
- Prototyped and evaluated DSP algorithms in Python to validate new product concepts and support research and development.
- Produced **10+** educational walkthrough videos explaining plugin functionality and underlying concepts to audio engineers and musicians.

#### MATHEMATICS SCIENCE TUTOR

Self Employed

In Person, Remote | Feb 2009 - May 2025

- Taught mathematics, physics and science to **70+ learners**, from primary school to master's level students across Russia, Israel, Germany and the UK.
- Built an educational YouTube channel about school mathematics, with **150+** videos, **50,000+ subscribers** and **4.5 million+ views**. Recieved the YouTube NextUP 2021 award.

#### RESEARCH ENGINEER

ITMO University

St. Petersburg, Russia | Mar 2011 - Jun 2017

- Conducted research in mathematical modelling within physical electrochemistry.
- Coordinated multiple research grants.
- Taught undergraduate laboratory classes in Numerical Methods in Programming.
- Authored **13+** publications and presented research at **20+** international conferences.

#### ONLINE COURSE PRODUCER AND CONSULTANT

Self Employed

In Person, Remote | Apr 2017 - Present

- Designed **15+ online learning programmes** for corporate and non-profit organisations.
- Authored **4** interactive B2B e-learning courses covering presentations and public speaking.
- Designed and facilitated **five** multi-day workshops on presentations and public speaking for corporate and government audiences as well as other educators.

### LEADERSHIP & COMMUNITY ENGAGEMENT

#### Volunteer Coordinator

- ADC Online (2025)
- CppOnline (2025)

#### Online Volunteer

- ADC (2023, 2024)
- CppOnline (2023, 2024)
- ACCU (2023, 2024)

#### FOUNDER & FACILITATOR – AUDIO DEVELOPER COMMUNITY

Remote | Nov 2024 - Present

- Founded a peer-learning community "Audio Builders Foundry" connecting aspiring audio developers to collaborate on portfolio projects.
- Facilitated the formation of project teams and established an ongoing technical study group.
- Created an environment that enabled participants to complete collaborative projects, including one presented as a poster at the Audio Developer Conference 2025 (online).
- Grew the initiative into an ongoing study group supporting continuous learning within the audio development community.